

Skills Development for Adolescent Girls

Investment case · East Java, Indonesia

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Contents

1 • Diagnostic — the size and shape of the gap

2 • Testing the relationships — regression analysis of what is associated with NEET

3 • Evidence & Theory of Change — what has worked in Indonesia, and the proposed programme

4 • Costing & ROI — component costs and benefit-cost ratio

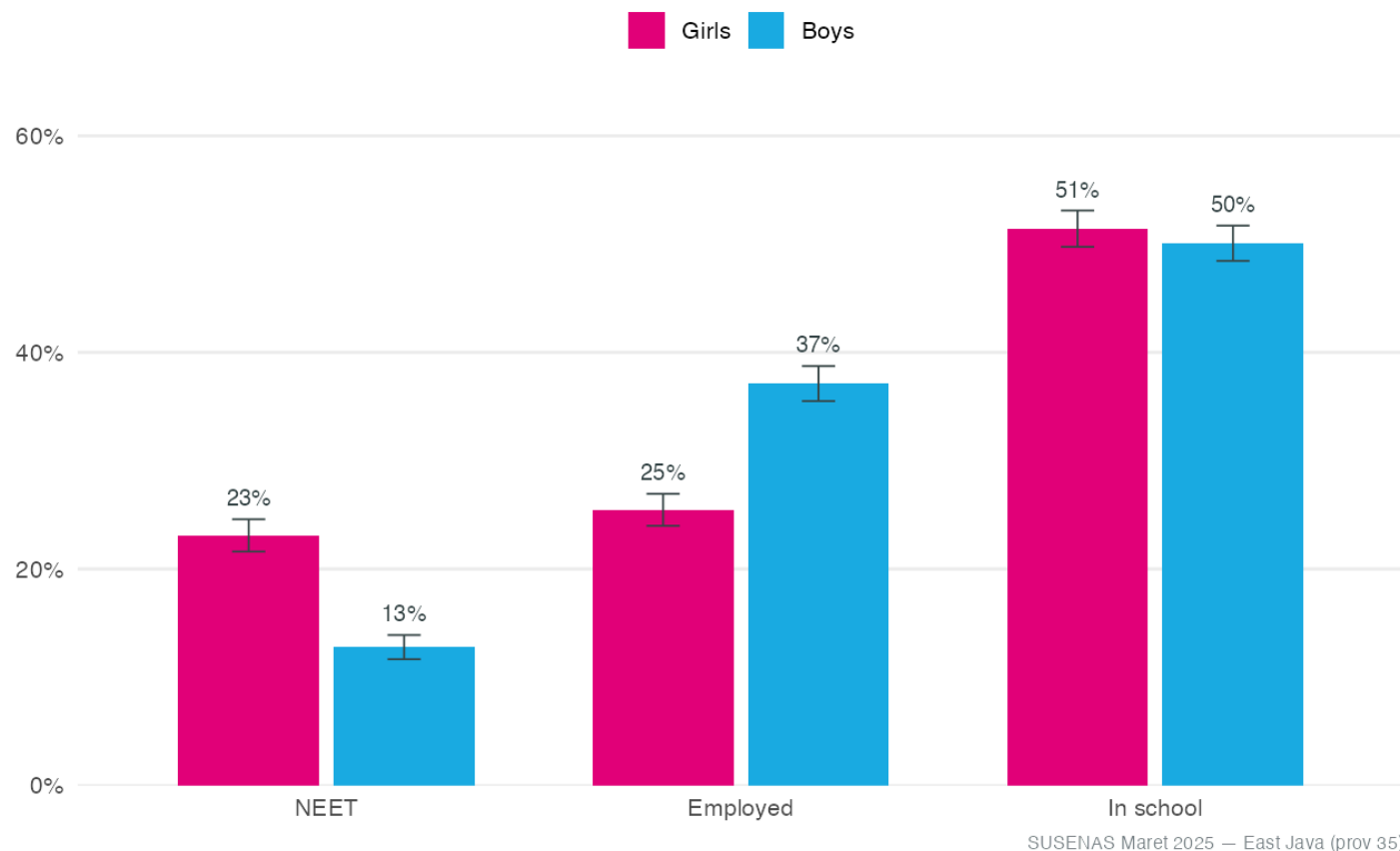
5 • Key Findings

Appendix — data, definitions, and methods

1 • Diagnostic

Girls, boys, and the school-to-work transition

Girls match boys in school — but carry ~2x the NEET rate and far lower employment



Girls 15–24 are slightly ahead of boys in school attendance (51% vs 50%), but their NEET rate is close to double (23% vs 13%) and their employment rate is 12 points lower (25% vs 37%). The difference appears at the transition out of school, not within it. SUSENAS Maret 2025, survey-weighted; 95% CIs.

Forgone lifetime productivity from the NEET gap

What East Java is losing right now

Forgone lifetime productivity from the 10.3pp girls-vs-boys NEET gap — a present cost of inaction, not a future benefit

Per excess-NEET girl (USD)

USD 14,143



Province-wide, today (USD millions)

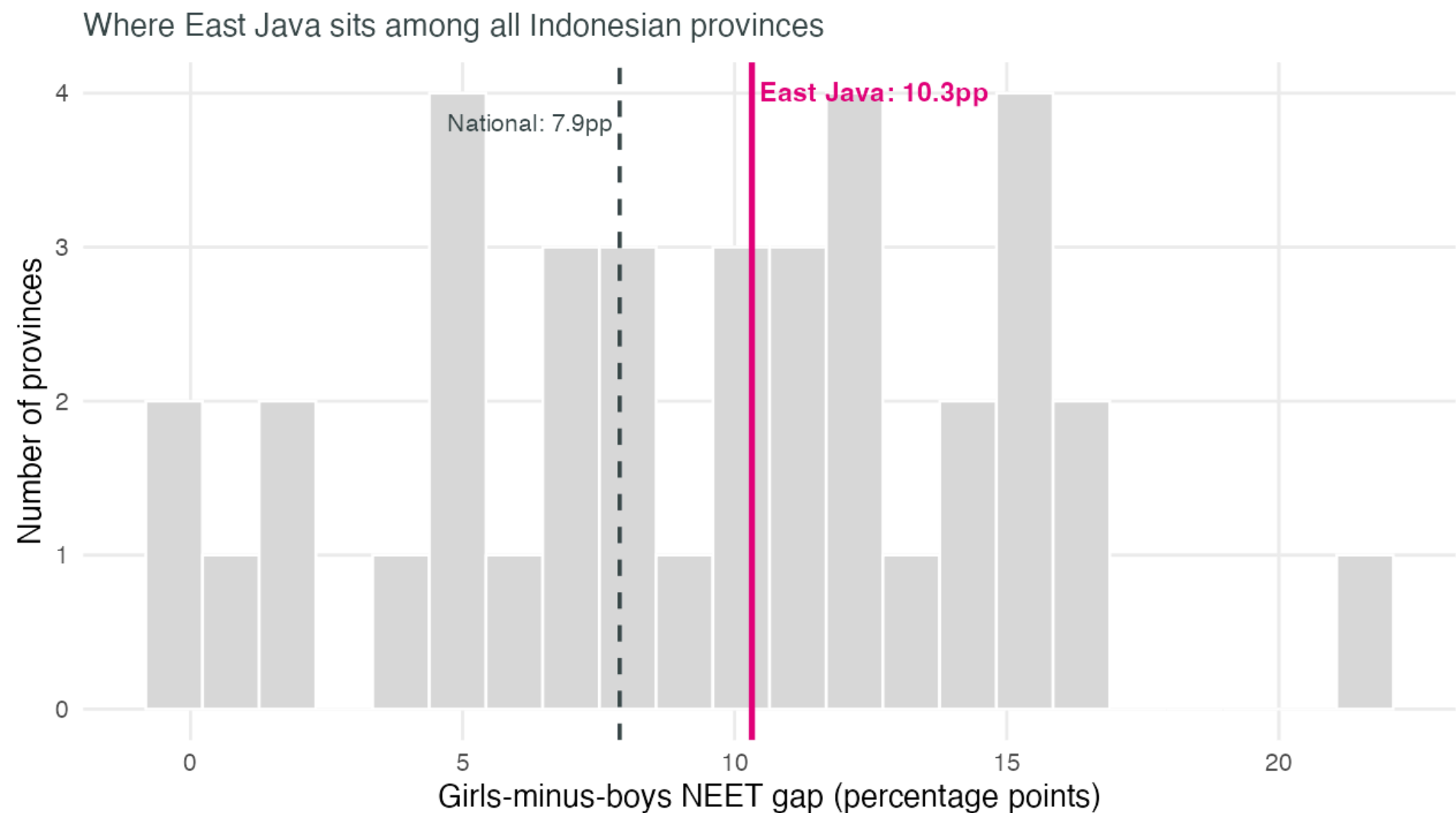
USD 4,077.0M



SUSENAS Maret 2025 — East Java (prov 35)

What this is: the number of girls who are NEET in excess of the boys' rate, multiplied by the earnings each would be expected to make over a working lifetime, discounted to present value.

The gap in national context

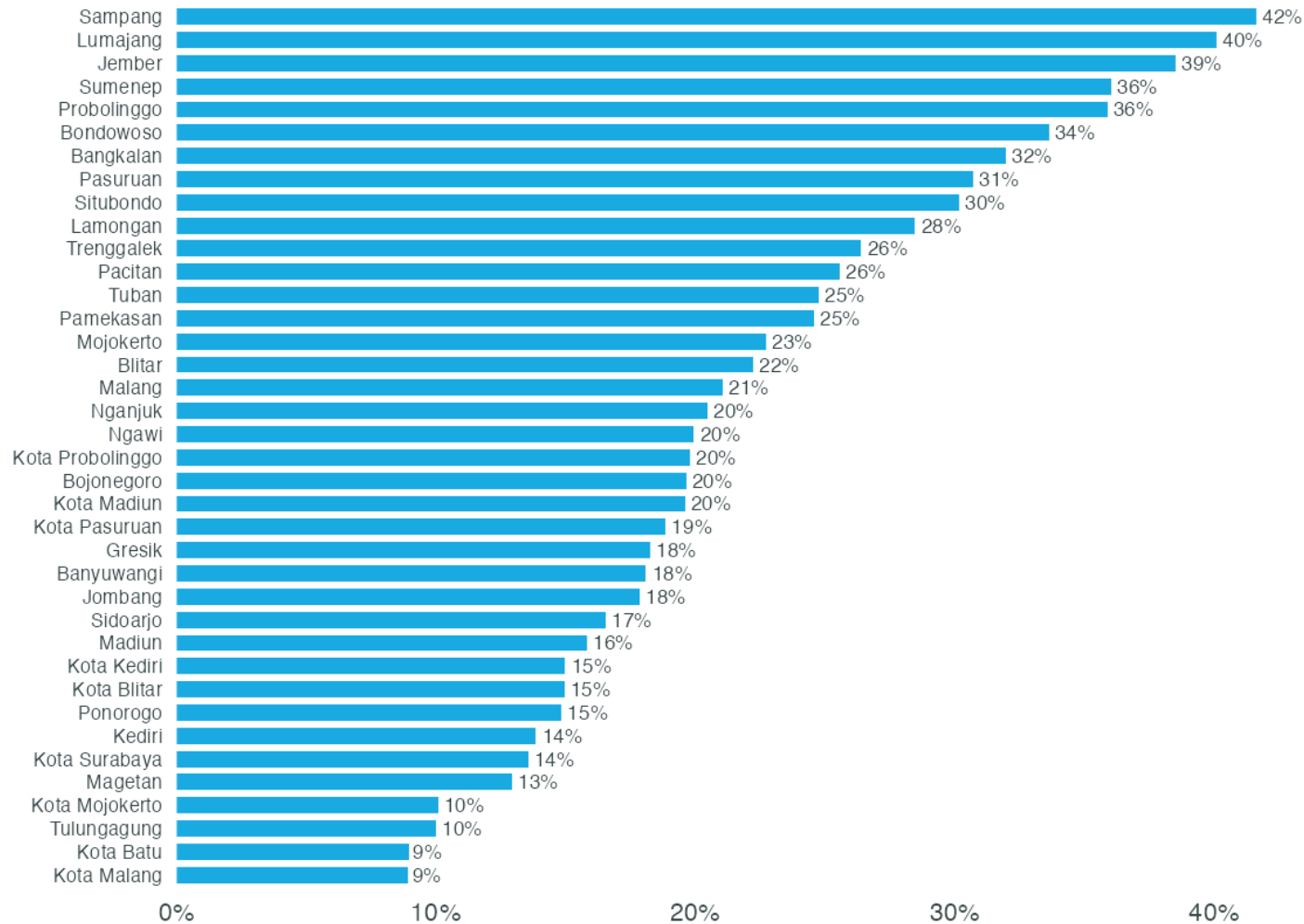


SUSENAS Maret 2025, all provinces, survey-weighted; adolescents 15-24

The same NEET gender gap computed for every Indonesian province from the same survey, showing where East Java sits relative to the national average.

NEET among girls, by district

Girls' NEET rate, by East Java district (≥15 sampled girls)



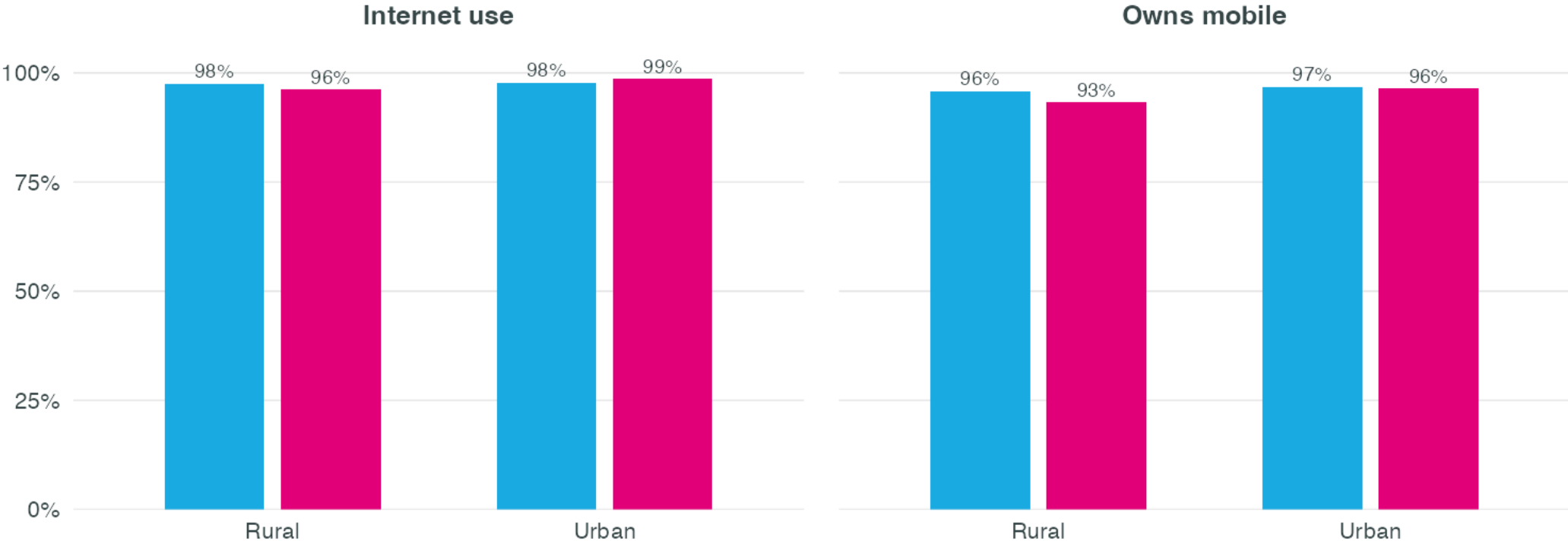
Source: SUSENAS Maret 2025 — East Java (prov 35). Weighted, adolescents 15–24.

SUSENAS Maret 2025, girls 15–24, survey-weighted; East Java districts with ≥15 sampled girls.

Digital access and use

Digital access and use, by sex and area

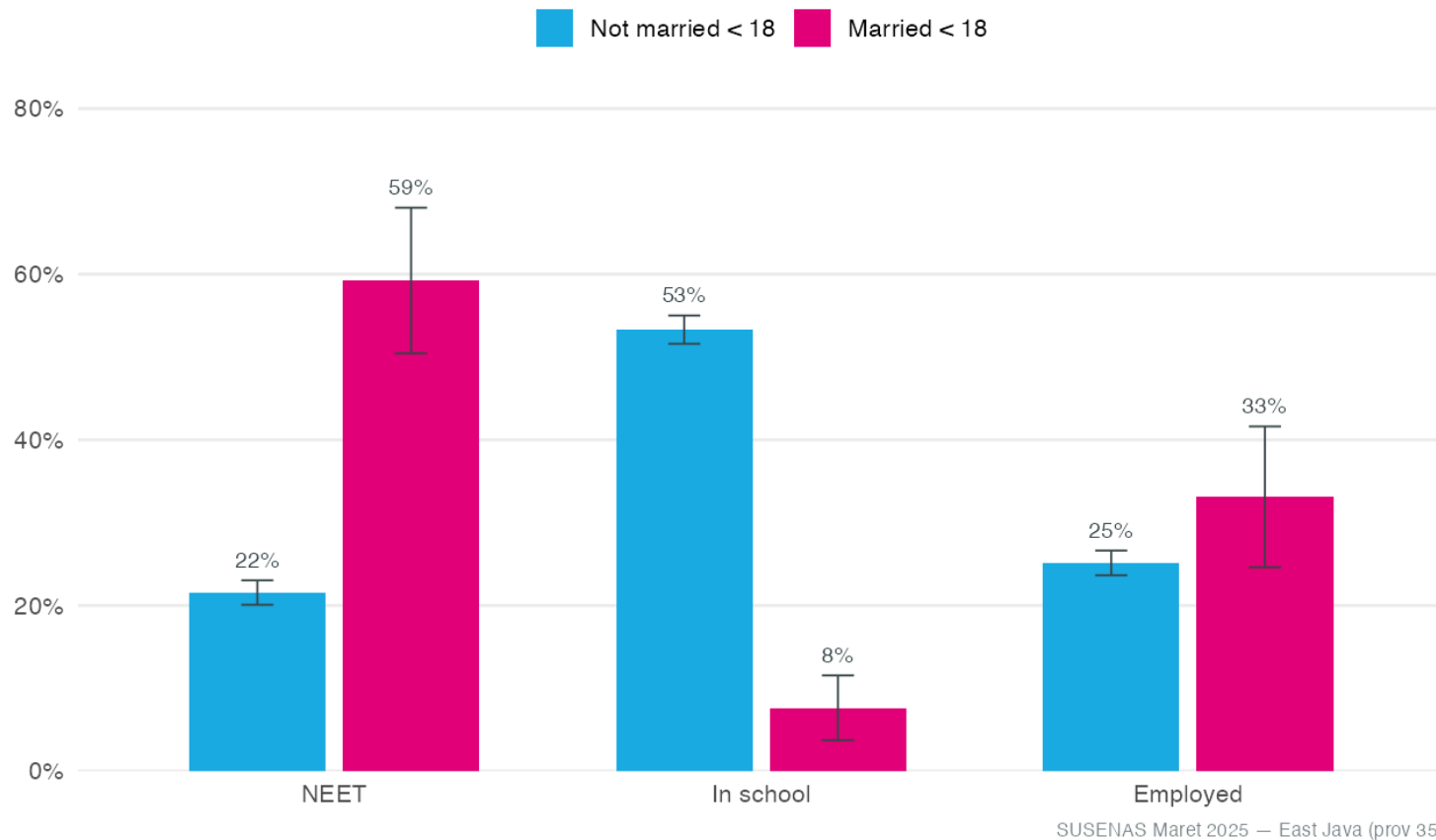
Boys Girls



Source: SUSENAS Maret 2025 — East Java (prov 35). Weighted, adolescents 15–24.

Child marriage, schooling, and labour-market outcomes

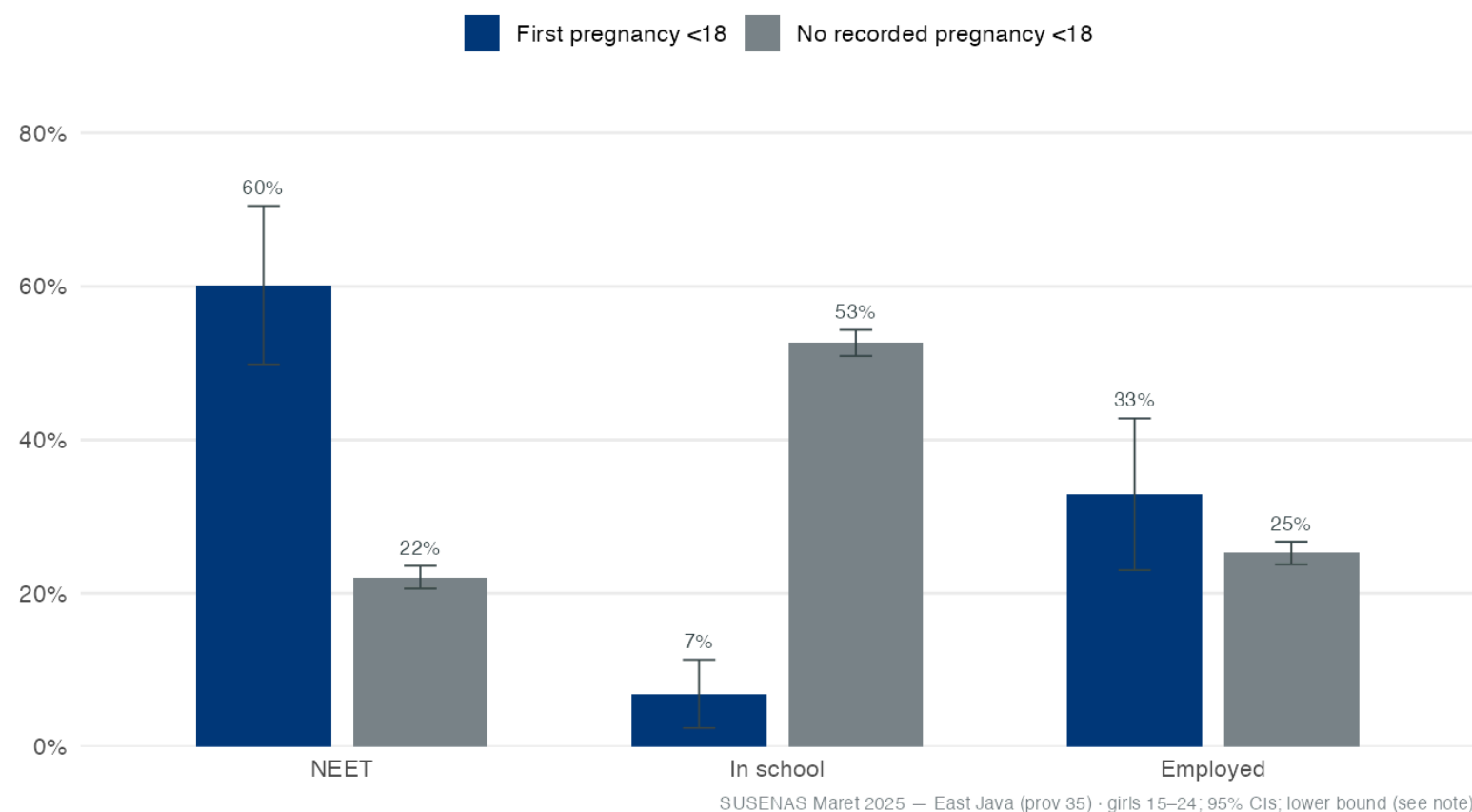
Girls married before 18 are far more likely to be NEET and out of school



Girls 15–24, survey-weighted (95% CIs). Girls who married before 18 have a NEET rate of 59% against 22% for those who did not, and 8% are still in school against 53%. Around 4% of East-Java girls 15–24 married before 18. **Association, not causation** — the direction is not identified, and most NEET girls did not marry before 18.

Early pregnancy, schooling, and labour-market outcomes

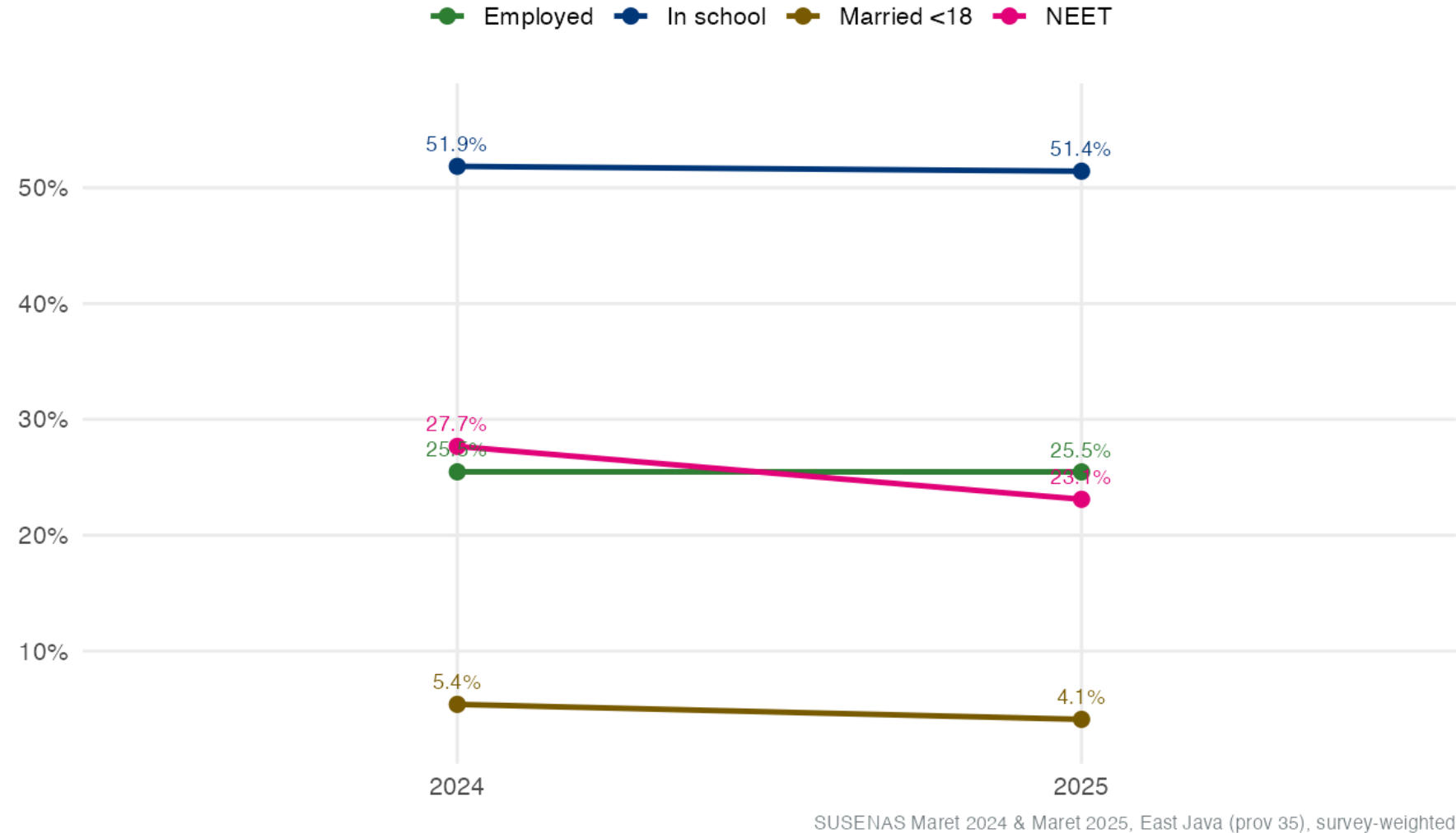
Girls with an early pregnancy are far likelier to be NEET and out of school



First pregnancy before 18 (SUSENAS Blok XV). SUSENAS records pregnancy history only for ever-married women, so these rates are a lower bound on the true prevalence.

Change between 2024 and 2025

Girls 15-24, East Java — one year of movement (SUSENAS Maret 2024 vs Maret 2025)

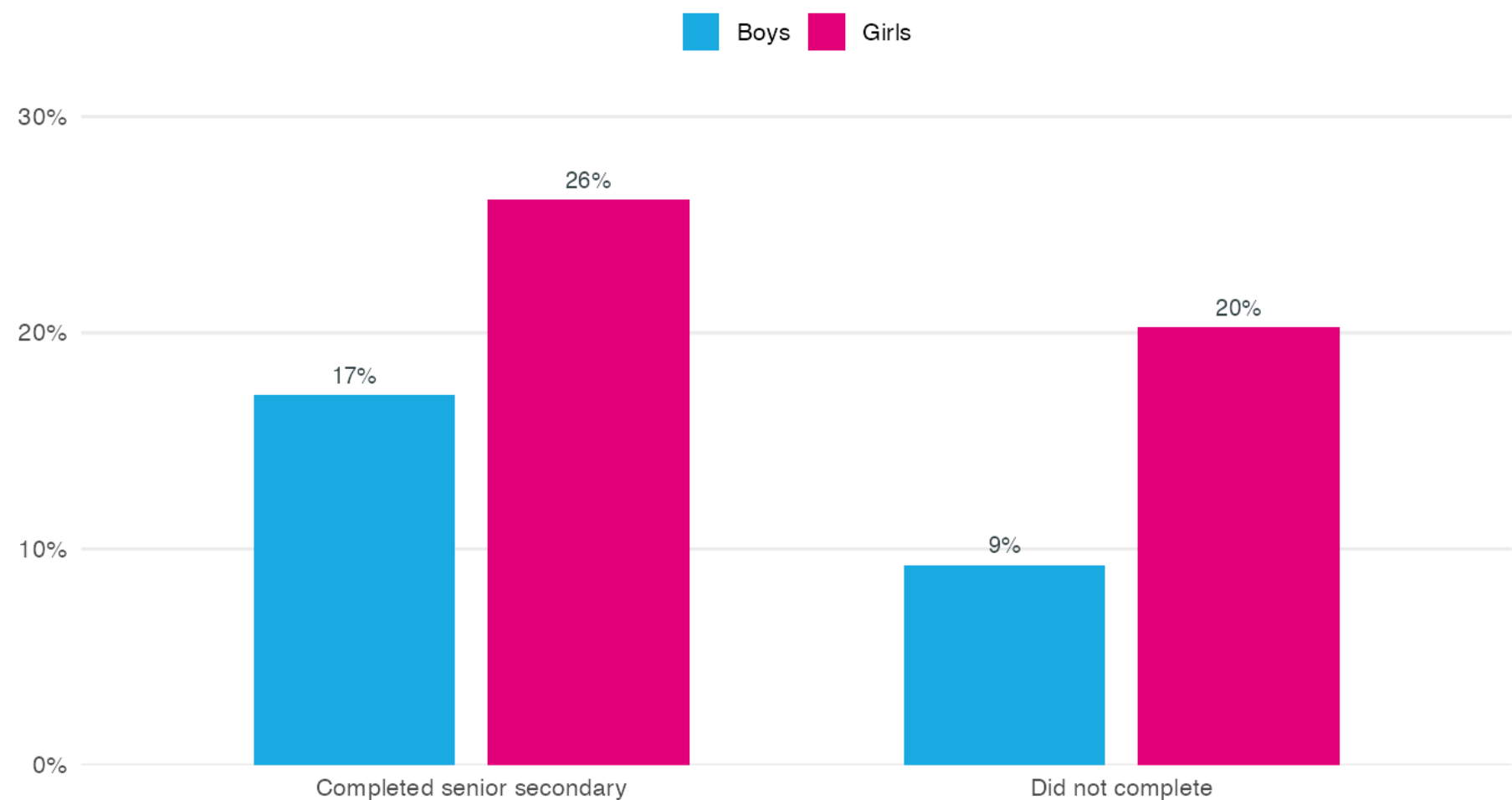


Two consecutive SUSENAS rounds for girls 15–24. Girls’ NEET fell 4.6 points and child marriage 1.3 points; school attendance and employment were essentially flat.

2 · Testing the relationships

Senior-secondary completion and NEET

Completing senior secondary is associated with substantially lower NEET, for both sexes



SUSENAS Maret 2025 — East Java (prov 35) · adjusted for age, rural (poverty control unavailable this run) — see completion_neet.csv

Regression results — marriage, pregnancy and exclusion

Hypothesis	Effect (pp)	SE	p
Child marriage → NEET	28.7	4.6	0.0000
Child marriage → in school	-27.5	3.0	0.0000
Child marriage → employment	-1.2	4.6	0.7942
Early pregnancy → NEET	29.6	5.4	0.0000
Early pregnancy → in school	-28.0	3.7	0.0000
Early pregnancy → employment	-1.5	5.4	0.7802
Early pregnancy ↔ child marriage (association)	97.4	0.4	0.0000

Linear-probability estimates, girls 15–24, survey-weighted, PSU-clustered. Effects in percentage points. Controls: age and urban/rural throughout, plus poverty where expenditure data is available. Robust to adding district fixed effects. Association, not causal identification.

3 · Evidence & Theory of Change

Evidence from Indonesia

Programme	Design	Result	Relevance
PKH conditional cash transfer (Priebe & Sumarto, 2025)	Quasi-experimental	Girls' junior-secondary enrolment +5.2pp; child marriage -3.5pp	Direct evidence a cash component reduces child marriage in Indonesia
PIP education assistance (Ulfa & Rezki, 2024)	Quasi-experimental	School attendance +11.4pp	Anchors the attendance effect used in the ROI model
PIP — girls' senior secondary (Caniago, 2021)	Quasi-experimental	Girls' senior-secondary enrolment +24.6pp	Senior-secondary transition, the stage this programme targets
Vocational training (Pritadrajati, 2022)	Quasi-experimental	Female LFPR +3.4pp; income +5.6%; BCR up to 22.4	Indonesian precedent for skills-to-work returns
NEET in Indonesia (Faizah, Isbah & Azca, 2025)	Qualitative	Norms, care duties and marriage drive girls' NEET	Explains the transition gap seen in Section 1

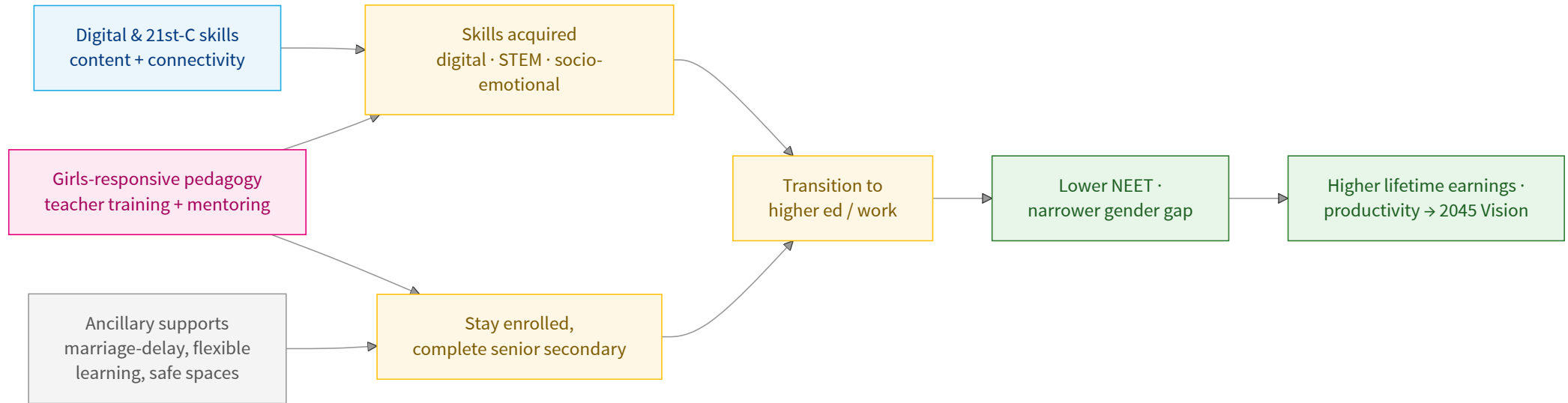
The ROI model in Section 4 is anchored on **PIP/PKH Indonesian effect sizes directly** — no cross-country transport discount is applied, because domestic evidence exists for this intervention type.

Comparable international evidence

Programme	Design	Girls / age	Focus	Key result	Fit for East Java
Adolescent girls' wellbeing — Bergström & Özler 2021	Review (104 studies)	Adolescent girls	Multi-sector interventions	Integrated packages beat single-sector ones	Synthesis backbone for the ToC
NEET in Indonesia — Faizah, Isbah & Azca 2025	Qualitative	Youth (gendered)	Drivers of girls' NEET	Non-education factors (norms, care, marriage) drive girls' NEET	Indonesia-specific NEET context
ELA girls' clubs — Bandiera 2018 (Sierra Leone)	RCT	Girls 12–25	Vocational + life skills	Enrolment +8.5pp; more income-generating activity	Closest Skills4Girls-in-Double-Track model
Female digital-employment bootcamp — Alzúa 2020 (Argentina/Colombia)	RCT	Women 18–50	Coding for employment	Large coding-skill gains	Core digital-skills → jobs arm
Irumi — Näslund-Hadley 2025 (Paraguay)	RCT	Pupils	Computational thinking	+0.09 SD CT	Structured CT pedagogy
Robotics & Coding — Gülen 2025 (Turkey)	RCT	Girls 14–16	Robotics / coding	Higher STEM-career orientation	Girls' STEM, secondary age
Samata — Prakash 2019 (India)	RCT	Poor girls 13–16	Secondary-transition support	More girls transition; norm shift	South Asia; senior-secondary transition
Room to Read GEP — Edmonds 2021 (India)	RCT	Girls	Life skills + mentoring	Dropout –3.5 to –4.1pp	Mentoring/pedagogy layer cuts dropout
Negotiating a Better Future — Ashraf 2025 (Zambia)	RCT	Girls ~14	Negotiation skills	+0.26 years schooling (10-yr)	Durable, cheap soft-skills layer
OLPC / Plan Ceibal — de Melo 2014 (Uruguay)	Quasi / review	Pupils	Laptops (hardware-led)	Learning ≈ null (–0.05 SD)	Caution: hardware alone fails → fund pedagogy

Effects as reported in each source. Used to inform programme design; the ROI effect size itself comes from the Indonesian evidence on the previous slide.

Theory of change



Embedded in Double Track / **Tambahan Keterampilan**. Measured against the ToR: cost per NEET averted, transition to higher education or work, BCR.

4 · Costing & ROI

Component costs

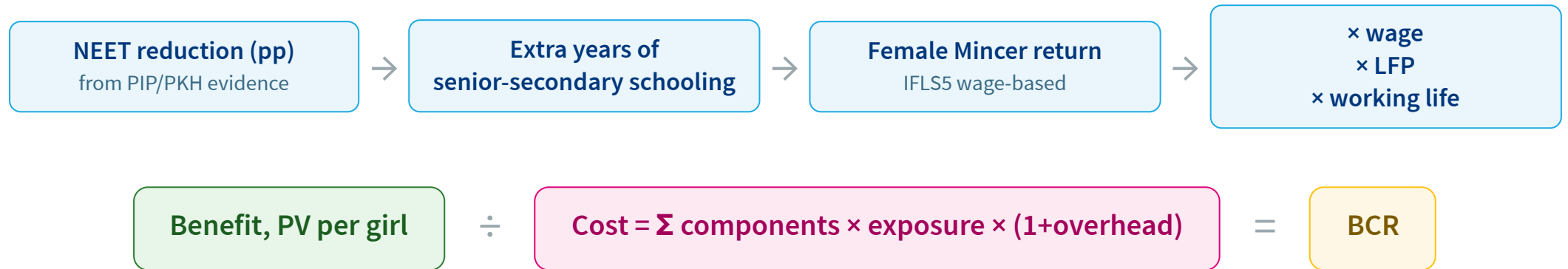
All figures are **USD per girl, per year of programme exposure**.

USD per girl per year — Low / Central / High

Component	Low	Central	High
Digital content & platform	5	10	15
Teacher training (annualised)	8	15	25
Mentoring & socio-emotional	5	10	15
School connectivity/devices (shared)	8	15	25
Ancillary: marriage-delay + flexible learning	4	8	14
TOTAL per girl/year (USD)	30	58	94

Source/method: Programme- and literature-benchmarked unit costs, using Indonesia-specific sources where available. Training and mentoring costs are annualised from comparable programmes; connectivity assumes shared labs rather than 1:1 devices. Ancillary activities reflect ToR guidance, and overheads follow UNICEF’s 10–20% range. Low/central/high estimates capture sourcing uncertainty. Replace with Double Track actuals once provincial financing data are available.

Benefit value chain



The effect chain runs left to right into the benefit; cost is computed independently, and the two combine into the BCR.

BCR and cost per NEET case averted

“NEET averted” is a count of girls, not a percentage point: the pp effect multiplied by girls exposed gives an expected number of cases prevented.

Scenario	Years edu gained	Benefit PV (USD/girl)	Cost (USD/girl)	NEET cases averted per exposed girl	BCR	Cost per full NEET case averted (USD)
Low	0.04	13	451	0.02	0.03	22,560
Central	0.15	233	200	0.05	1.17	4,002
High	0.32	1197	66	0.08	18.13	825

The Low scenario carries the null effect (its confidence interval includes zero), so its BCR is not defined — the return is conditional on a real effect, to be confirmed at pilot.

What the BCR counts

Counted

Higher schooling → lifetime earnings premium for girls moved out of NEET, via the female Mincer return applied over the discounted working life.

Excluded (so the BCR is a lower bound)

Intergenerational gains · delayed child marriage · health and productivity spillovers · fiscal returns · broader empowerment

5 • Key Findings

Key findings

1. Girls 15–24 match boys in school attendance but have close to double the NEET rate (23% vs 13%) and 12 points lower employment. The gap is at the school-to-work transition.
2. That gap carries a measurable present cost in forgone lifetime productivity, quantified in Section 1 and benchmarked against the national average.
3. Senior-secondary completion is associated with substantially lower NEET for both sexes — the mechanism the programme targets.
4. Child marriage and early pregnancy are each strongly associated with higher NEET and lower school attendance, though they account for a minority of NEET cases.
5. Girls' NEET fell 4.6 points and child marriage 1.3 points between 2024 and 2025, while attendance and employment held flat.
6. Indonesian evidence (PKH, PIP, vocational training) supports both the skills component and a cash-transfer component, and anchors the ROI model without cross-country extrapolation.
7. Open items pending external data: disability (2023 SUSENAS), Sakernas wage microdata, higher-education field of study (PDDikti), and the 2025 expenditure module for poverty analysis.

Appendix

Data and sample

Source. SUSENAS Maret 2025, East Java (province 35) — 98,738 individuals; adolescents 15–24: **12,599** (6,060 girls). Survey-weighted, PSU/strata-clustered ([WEIND](#), [PSU](#), [STRATA](#)). SUSENAS Maret 2024 retained for the year-on-year comparison.

Definitions:

- **NEET (15–24)** = not currently attending school ([R611](#)) and not working — built from the activity block ([R703_A](#) worked, [R703_B](#) schooled, [R704](#) temporarily absent from a job).
- **Sex** [R405](#) · **age** [R407](#) · **married before 18** from age at first marriage ([R409](#)) · **early pregnancy** from age at first pregnancy (Blok XV, ever-married women only).
- **Educational attainment** from highest certificate held ([R615](#)).

Note on variable numbering: the 2025 questionnaire renumbered several variables relative to 2024 (school participation R610→R611, certificate R614→R615, internet R805→R808, pregnancy block 1501→1401) and replaced the single main-activity question with a multi-select block. The 2024 and 2025 cleaning scripts are therefore separate files.

Data availability

Indicator	Status
NEET, schooling, employment, digital access	Available (2024 and 2025)
Child marriage, early pregnancy	Available; pregnancy is ever-married women only (lower bound)
Expenditure, wealth quintiles, official poverty	Not in the 2025 merged file — the consumption module is a separate BPS release, not yet obtained
Wages	Not collected in SUSENAS; IFLS5 used for the Mincer return
Disability	Not in the 2024 or 2025 KOR extract; available in SUSENAS 2023
Higher-education field of study	Not collected in SUSENAS; PDDikti is the source
Programme costs	Pending from the provincial government

The Mincer specification

Returns to schooling are estimated as:

$$\ln(Y_i) = \beta_0 + \beta_1 S_i + \beta_2 S_i^2 + \beta_3 \text{Age}_i + \beta_4 \text{Age}_i^2 + \beta_5 \text{Urban}_i + \varepsilon_i$$

- Y_i — monthly wage (IFLS5 earners 20–60)
- S_i — years of schooling, from highest certificate attained
- β_1 — the return per additional year of schooling
- β_2 — the quadratic term; positive and significant indicates returns that rise with schooling level
- Estimated weighted, with robust standard errors

A consumption-based version of the same specification (using SUSENAS per-capita expenditure in place of wages) is available once the 2025 expenditure module is obtained; the 2024 estimate was +5.6%/yr for women, +5.2%/yr for men, with significant convexity.